Semantic Knowledge Base Construction

LEVEL 3: Infer Relations

[SAT] → (FK: CSDCode) → [Schools]

1:1 Match

LEVEL 2: Analyze Semantics

Column Semantics:

- AvgScrMath: Average mathematics score

Table Semantics:

- Table: SAT_Scores Academic performance metrics (Read / Write / Math)

LEVEL 1: Extract Metadata

Domain Context:

- Topic: Public School Academic & Socioeconomic DB

Schema Statistics:

- Table: SAT_Scores
- Primary Key: CSDCode Tag: High Entropy

Reasoning-Traceable Generation & Validation

Context & Constraints

- Schema & Knowledge Base
- Scenario: Ranking

LLM →

Synthesis Explanation

Step 1: Goal Analysis
Identify one school with Max(AvgScrMath)

Step 2: Resource Mapping
Table: 'satscores' (Contains SAT)

Step 3: Strategy Formulation
SQL: Select, Order by ASC, Limit 1;

ASC → DESC

Diagnose

LLM ↓

Question:
Which school has the highest math score?

SQL:

```
SELECT sname FROM satscores
ORDER BY AvgScrMath ASC LIMIT 1;
```

Detailed Process

